Homework 22

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. You are developing a parallel simulation where each process needs to send its local state to a master node. The state consists of:
   - A `double` representing the velocity.
   - A `float` representing the mass.
   - An `int` array of size 3 representing the 3D coordinates (x, y, z).

   A. Define a new MPI derived datatype using `MPI_Type_create_struct`.
   B. Explain why you must use `MPI_Get_address` to find the displacements of each variable instead of simply using the `sizeof()` operator.
   C. Write the `MPI_Type_commit` and `MPI_Type_free` sequence required to use and clean up this type.

2. Consider the following code used to time a parallel computation across a communicator comm:

```
MPI_Barrier(comm);
start = MPI_Wtime();
/ Computation Block /
end = MPI_Wtime();
local_elapsed = end - start;
MPI_Reduce(&local_elapsed, &elapsed, 1, MPI_DOUBLE, MPI_MAX, 0, comm);
```

   A. Explain the specific purpose of calling `MPI_Barrier(comm)` before taking the start time.
   B. Why is `MPI_MAX` used in the `MPI_Reduce` function instead of `MPI_SUM` or `MPI_MIN` when calculating the total elapsed time of the parallel block?
   C. If the barrier was removed, how would the reported elapsed time on the master node be affected?.

- **Welcome: Sean**
  - LogOut